User Guide – SmartDetect AI: Voice Deepfake Detection System

Project: AI Voice Deepfake Detection
Authors: Karina Pisarenko, Nofar Horada
Institution: Final Project – Information Systems

1. Introduction

The AI Voice Deepfake Detection System is a web-based application developed to identify whether an uploaded audio recording is authentic or AI-generated (deepfake).

The system is based on a fine-tuned WavLM deep learning model and provides real-time analysis through an interactive dashboard developed using Streamlit.

The application was designed to provide a simple and intuitive interface, allowing users to upload an audio recording and receive an immediate prediction together with the model's confidence level and additional analysis information.

Audio Upload
↓
Preprocessing
↓
WavLM Model
↓
Prediction
↓
Results Display
↓
Database Storage

2. System Requirements

The application is web-based and does not require any software installation

Requirements:

- Internet connection

- Modern web browser (Google Chrome, Microsoft Edge, Safari or Firefox)
- Supported audio formats: WAV (.wav), MP3 (.mp3), FLAC (.flac), OGG (.ogg), and M4A (.m4a).

3. Accessing the System

The dashboard can be accessed using the following URL:

Dashboard URL:

https://voice-deepfake-detection-airrw9l4kv2pqobhb5akax.streamlit.app/

After opening the link, the main dashboard will appear.

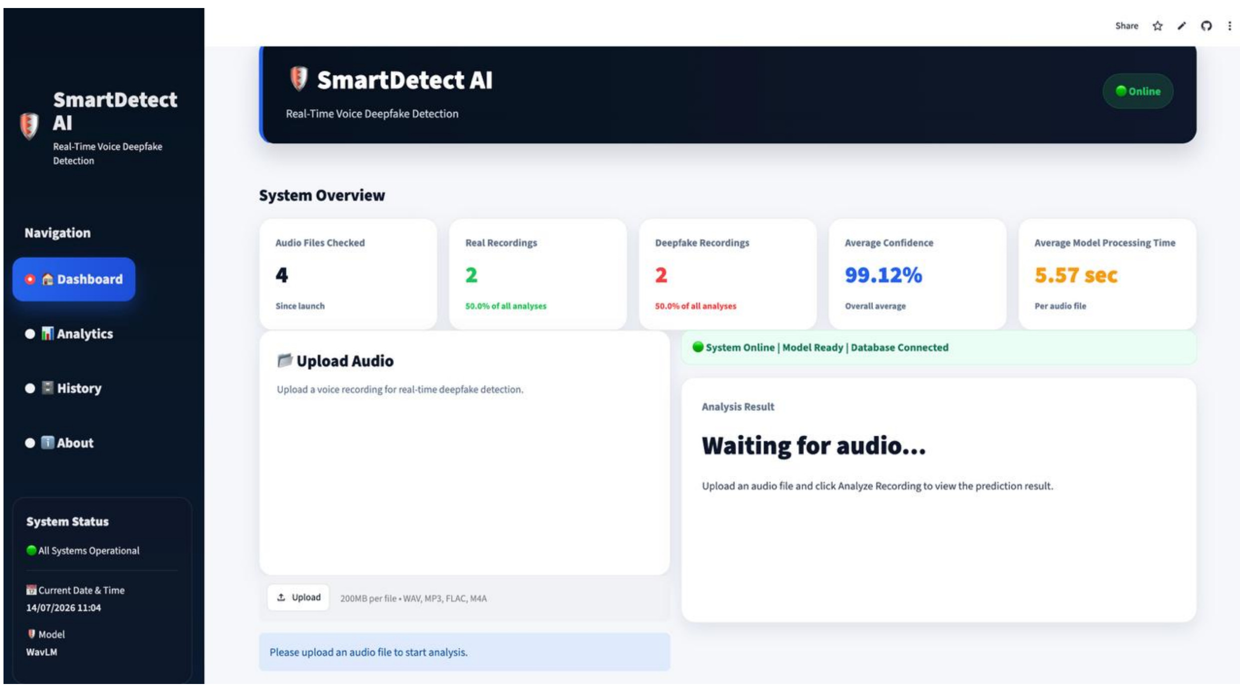


Figure 1. Main dashboard interface

4. Dashboard Overview

The dashboard consists of four main pages(Dashboard, Analytics , History, About) , each designed to provide a different aspect of the system's functionality.

1. **Upload Audio –** allows the user to upload an audio recording for analysis.
2. **Analyze Recording –** starts the deepfake detection process using the trained WavLM model.
3. **Detection Results –** displays the prediction generated by the model.

Possible outputs:

- Real
- Spoof Detected

4. **Confidence Score –** displays the prediction confidence as a percentage.
5. **Risk Level –** provides an estimation of the detected risk level.

   Possible values:

   - Low
   - Medium
   - High
6. **Audio Information –** displays information regarding the uploaded recording, including:
   - Recording duration
   - Processing time
   - Model used
7. **History** – allows users to review previously analyzed recordings and their corresponding detection results.
8. **Analytics** – presents statistical summaries and visualizations based on previous analyses.
9. **About** – provides information about the project, development team, model performance, system configuration, and application workflow.

5. Using the System

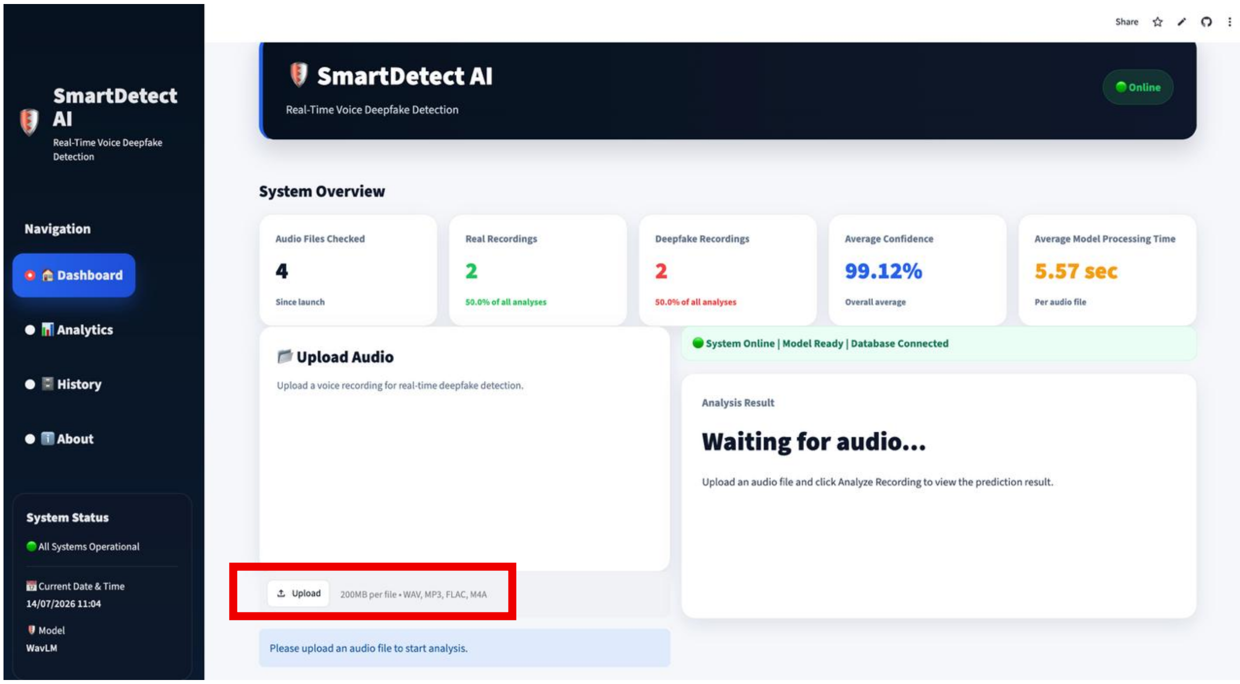
Follow the steps below to analyze an audio recording.

Step 1

Open the dashboard using the provided web link.

Step 2

Click **Upload**.

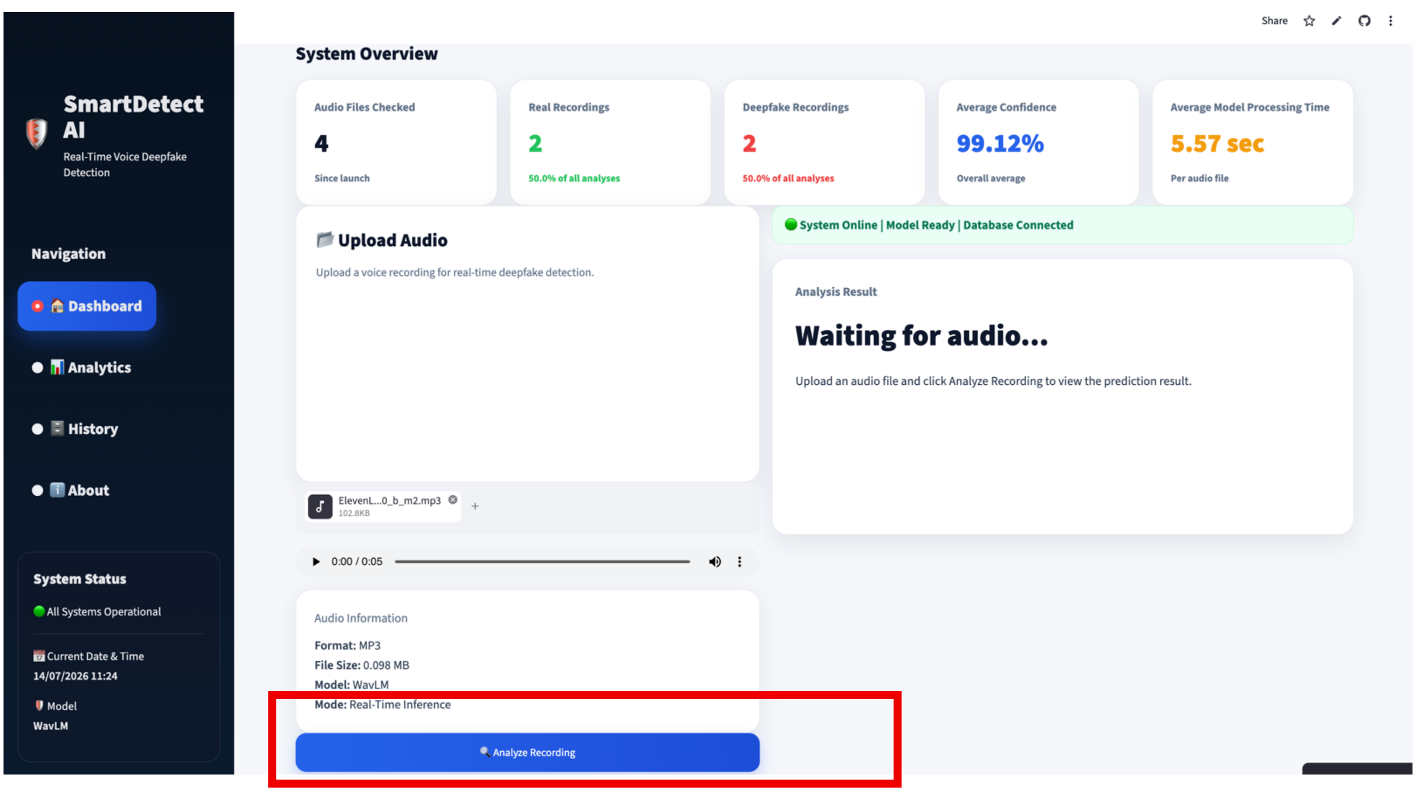


Step 3

Select the audio recording to be analyzed.

Step 4

Click **Analyze Recording**.

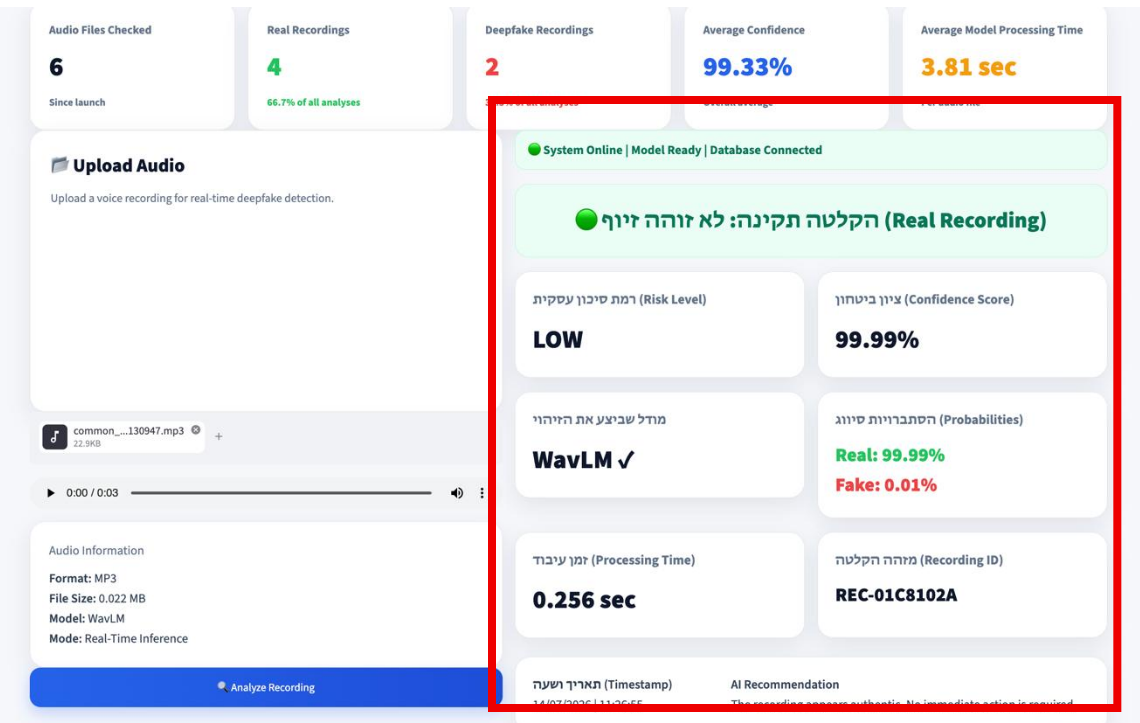


The system will automatically:

- preprocess the audio
- load the trained WavLM model
- classify the recording
- calculate confidence scores
- display the prediction

Step 5

Review the analysis results displayed on the screen.



6. Understanding the Results

After the analysis is completed, the dashboard displays several indicators describing the prediction results.

Prediction – represents the final decision of the AI model.

Possible values:

Real – the recording is classified as authentic.

Spoof Detected – the recording is classified as AI-generated or manipulated.

Confidence – represents the probability associated with the selected prediction. Higher confidence indicates that the model is more certain about its decision.

Risk Level – provides a simplified interpretation of the prediction confidence.

Risk Level	**Description**
Low	Recording is likely authentic
Medium	Suspicious recording requiring further verification
High	High probability of AI-generated speech

Processing Time – displays the time required to complete the analysis.

7. Analytics Page

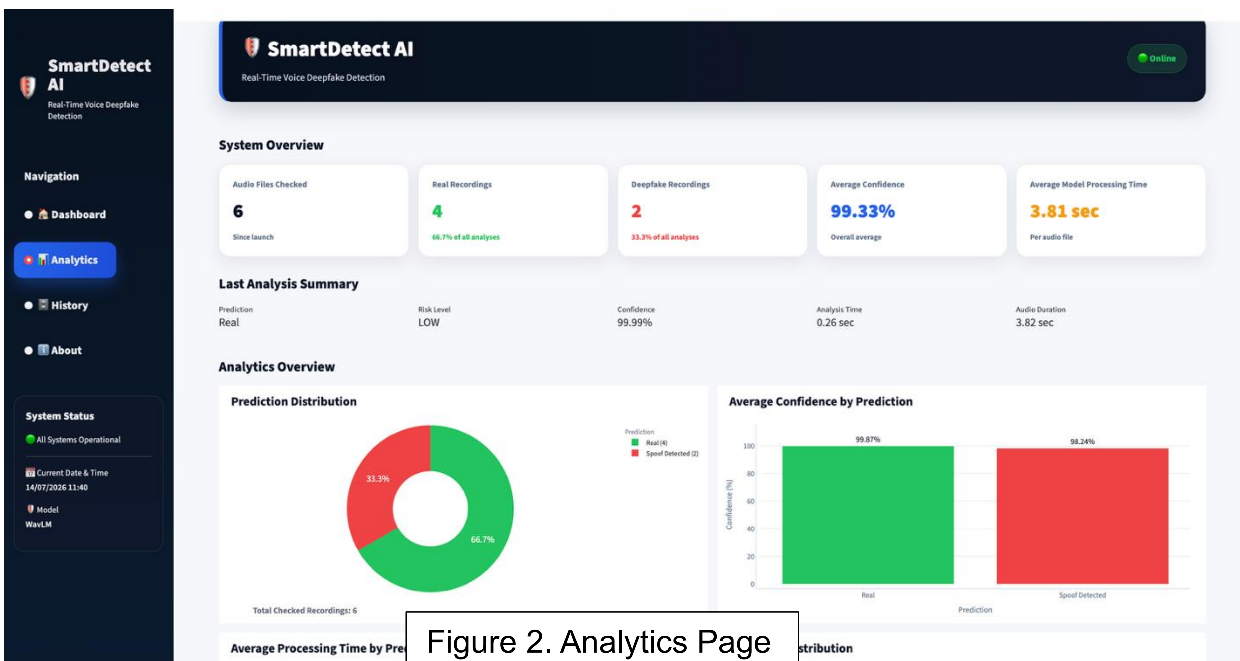


Figure 2. Analytics Page

The **Analytics** page provides an overview of all analyses performed using the system. It summarizes previous detection results through key performance indicators and interactive visualizations, enabling users to monitor system activity and identify overall trends.

The page includes the following sections:

System Overview

Displays key summary statistics collected since the application was launched:

- **Audio Files Checked –** total number of audio recordings analyzed by the system.
- **Real Recordings** – number of recordings classified as authentic.
- **Deepfake Recordings** –number of recordings classified as AI-generated.
- **Average Confidence** – average confidence score across all analyses.
- **Average Model Processing Time** – average time required to analyze an audio recording.

Last Analysis Summary

Displays the results of the most recently analyzed recording, including:

- Prediction
- Risk Level
- Confidence Score
- Analysis Time
- Audio Duration

Analytics Overview

The lower section of the page presents four analytical charts:

- **Prediction Distribution** – displays the proportion of recordings classified as Real and Spoof Detected.
- **Average Confidence by Prediction** – compares the average confidence score for each prediction category.
- **Average Processing Time by Prediction** – shows the average inference time for authentic and spoofed recordings.
- **Risk Level Distribution** – displays the number of recordings classified into each risk level.

These visualizations provide users with a comprehensive overview of system performance and previously analyzed recordings.

8. History Page

The **History** page stores every analysis performed through the dashboard.

For each analyzed recording, the following information is displayed:

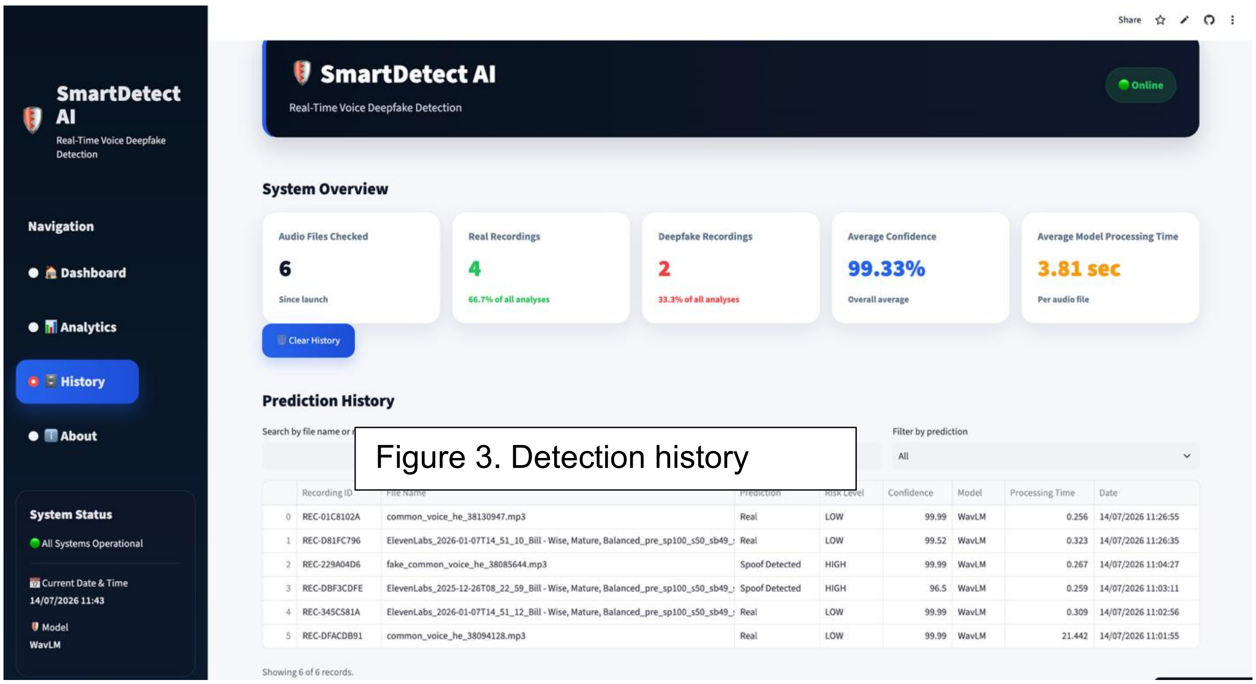


Figure 3. Detection history

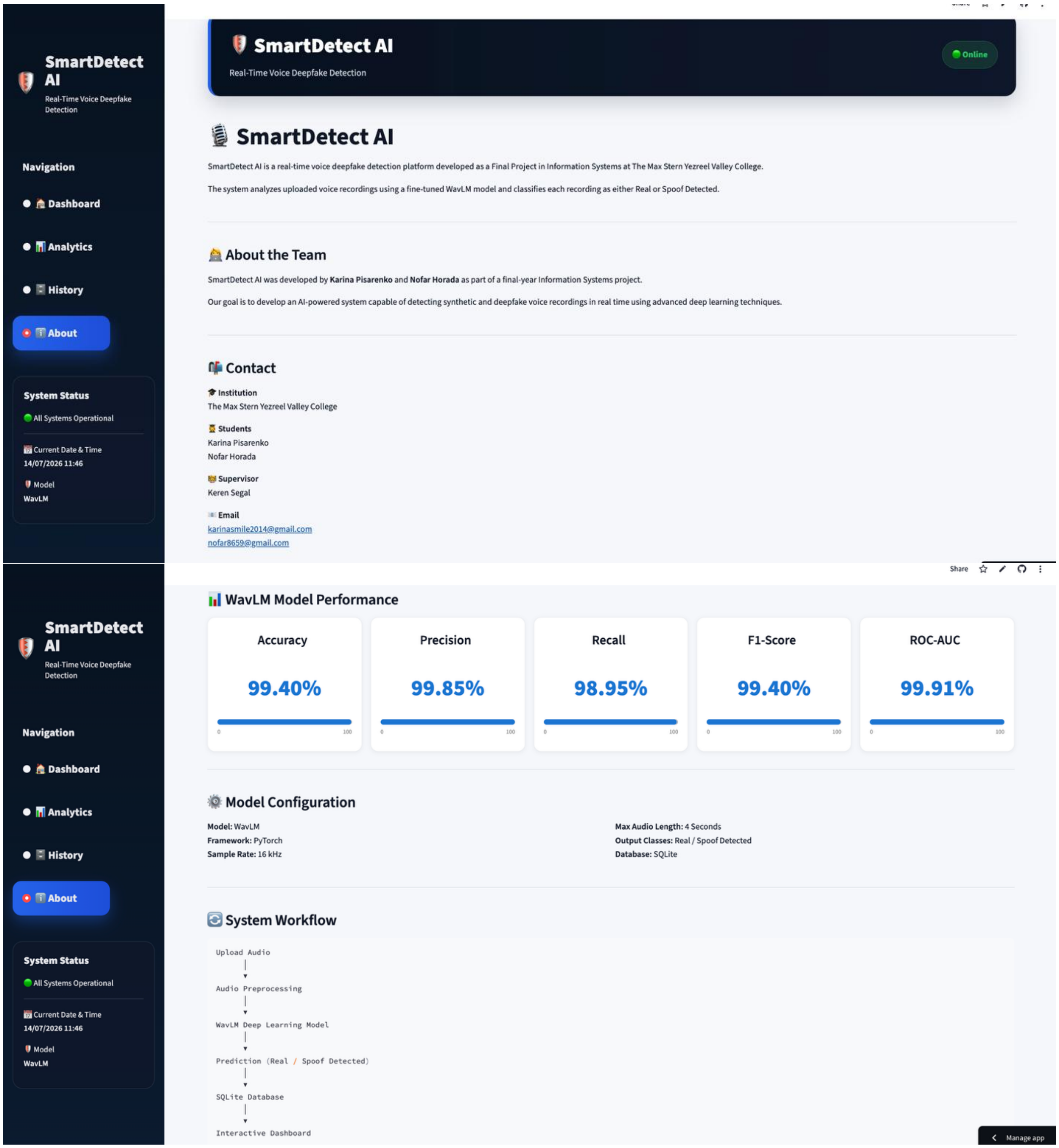
- Date and time of analysis
- Prediction result
- Confidence score
- Risk level
- Audio duration
- Processing time
- Model used

The history enables users to review previous analyses without repeating the detection process and serves as a record of all analyses performed through the dashboard.

9. About Page

The **About** page provides general information about the project and the development team.

The page includes the following sections:



Project Description

Provides a brief overview of the SmartDetect AI platform, describing its purpose and explaining that the system detects AI-generated voice recordings using a fine-tuned WavLM deep learning model.

About the Team

Introduces the project team and presents the overall objective of the project.

Contact Information

Displays general project information, including:

- Institution
- Students
- Academic supervisor
- Contact email

Model Performance

Presents the evaluation results of the selected WavLM model using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Model Configuration

Summarizes the technical configuration of the deployed model, including:

- Model name
- Deep learning framework
- Sampling rate
- Maximum supported audio length
- Output classes
- Database used by the system

System Workflow

The system performs the following operations after an audio recording is uploaded:

1. Upload audio file.
2. Audio preprocessing .
3. Loading the trained WavLM model.
4. Audio classification (Real or Spoof Detected).
5. Confidence score calculation.
6. Result presentation in the dashboard.
7. Saving the analysis results to the SQLite database.

This page provides users with both technical and descriptive information regarding the developed system.

10. Limitations

The current version of the system has several limitations:

- The model was trained and evaluated using a combination of public anti-spoofing English datasets together with a custom Hebrew dataset developed as part of this project.
- Performance may vary when analyzing recordings in unseen languages or with unfamiliar accents.
- The system currently supports binary classification (Real or Spoof Detected).

11. Troubleshooting

Problem	**Solution**
Audio file cannot be uploaded	Verify that the selected file is in a supported audio format.
First analysis takes longer than expected	During the first execution, the trained model is downloaded and loaded into memory. This process may take several minutes.
Dashboard does not load	Refresh the browser and verify the internet connection.

Problem	Solution
Prediction is not displayed	Upload the recording again and repeat the analysis.

12. Authors

Students

- Karina Pisarenko
- Nofar Horada

Academic Supervisor

- Dr. Keren Segal